

Set 9: Online Learning Activity Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To import an online-learning dataset into R, analyse quiz performance with a histogram, boxplot, bubble scatter plot and a smoothed monthly trend, and to build a filtered interactive dashboard in Tableau.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
# Create the CSV file (as provided) and import it into R
raw <- "Student_ID,Gender,Age,Course,Study_Time,Videos_Watched,Quiz_Score,Login_Date
L01,Male,20,R,3.5,12,78,2025-01-05
L02,Female,22,R,4.2,15,85,2025-01-05
L03,Male,19,SQL,2.0,8,65,2025-02-08
L04,Female,21,R,5.0,18,92,2025-02-08
L05,Male,23,R,2.5,9,70,2025-03-12
L06,Female,20,SQL,4.0,14,88,2025-03-12"
writeLines(raw, "Online_Learning_Activity.csv")

learn <- read.csv("Online_Learning_Activity.csv", stringsAsFactors = TRUE)
```

Q1. Import, Histogram and Boxplot of Quiz_Score

Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

R Program:

```
# Q1. Import, histogram of Quiz_Score, boxplot of Quiz_Score by Course
str(learn)
cat("\nSummary of Quiz_Score:\n"); print(summary(learn$Quiz_Score))
cat("Standard deviation:", round(sd(learn$Quiz_Score), 2), "\n")
cat("\nQuiz_Score by Course:\n")
print(aggregate(Quiz_Score ~ Course, data = learn,
  FUN = function(x) c(n = length(x), mean = round(mean(x), 1),
    median = median(x), min = min(x), max = max(x))))

p_hist <- ggplot(learn, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 10, boundary = 60, fill = "#0072B2", colour = "white") +
  geom_vline(xintercept = mean(learn$Quiz_Score), linetype = "dashed", colour = "#D55E00") +
  annotate("text", x = mean(learn$Quiz_Score) + 1, y = 2.6, hjust = 0, colour = "#D55E00",
    label = paste("Mean =", round(mean(learn$Quiz_Score), 1))) +
  scale_x_continuous(breaks = seq(60, 100, 10), limits = c(60, 100)) +
```

```

scale_y_continuous(breaks = 0:3) +
labs(title = "Histogram of Quiz Scores", x = "Quiz Score", y = "Number of Students") +
theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
ggsave("s9_q1_hist.png", p_hist, width = 6.5, height = 4, dpi = 150, bg = "white")

p_box <- ggplot(learn, aes(x = Course, y = Quiz_Score, fill = Course)) +
  geom_boxplot(width = 0.5, alpha = 0.7, outlier.shape = NA) +
  geom_point(position = position_jitter(width = 0.08, height = 0, seed = 1), size = 3) +
  scale_fill_manual(values = c(R = "#0072B2", SQL = "#E69F00")) +
  labs(title = "Quiz Score by Course", x = "Course", y = "Quiz Score") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"), legend.position = "none")
ggsave("s9_q1_box.png", p_box, width = 6.5, height = 4, dpi = 150, bg = "white")

```

Console Output:

```

'data.frame': 6 obs. of 8 variables:
 $ Student_ID   : Factor w/ 6 levels "L01","L02","L03",...: 1 2 3 4 5 6
 $ Gender       : Factor w/ 2 levels "Female","Male": 2 1 2 1 2 1
 $ Age          : int  20 22 19 21 23 20
 $ Course       : Factor w/ 2 levels "R","SQL": 1 1 2 1 1 2
 $ Study_Time   : num  3.5 4.2 2.5 2.5 4
 $ Videos_Watched: int  12 15 8 18 9 14
 $ Quiz_Score   : int  78 85 65 92 70 88
 $ Login_Date   : Factor w/ 3 levels "2025-01-05","2025-02-08",...: 1 1 2 2 3 3

Summary of Quiz_Score:
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
65.00  72.00   81.50   79.67  87.25   92.00
Standard deviation: 10.6

Quiz_Score by Course:
  Course Quiz_Score.n Quiz_Score.mean Quiz_Score.median Quiz_Score.min
1      R           4.0           81.2           81.5       70.0
2      SQL          2.0           76.5           76.5       65.0
  Quiz_Score.max
1           92.0
2           88.0

```

Output:

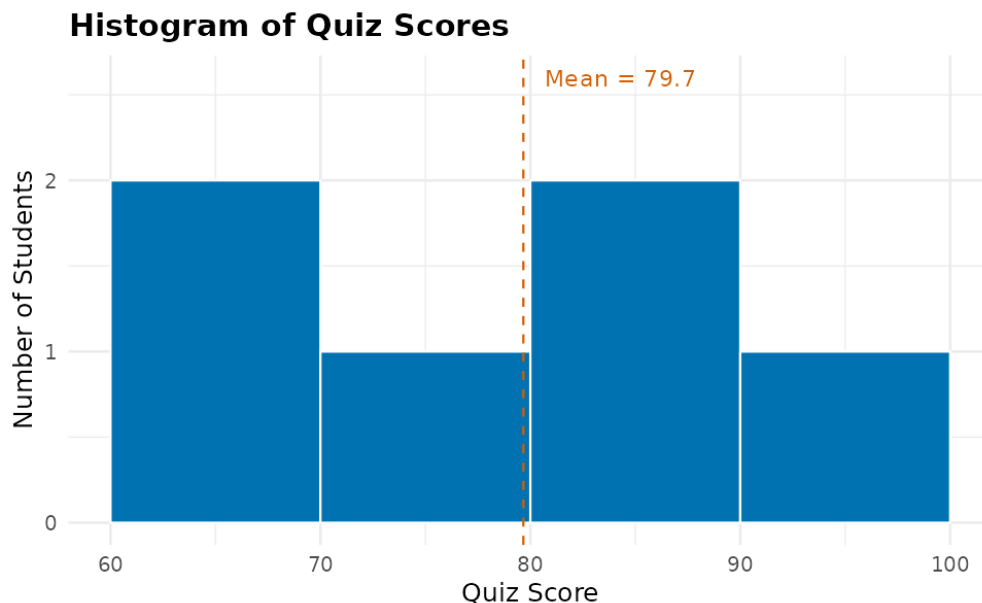


Figure 1: Histogram of quiz scores

Output:

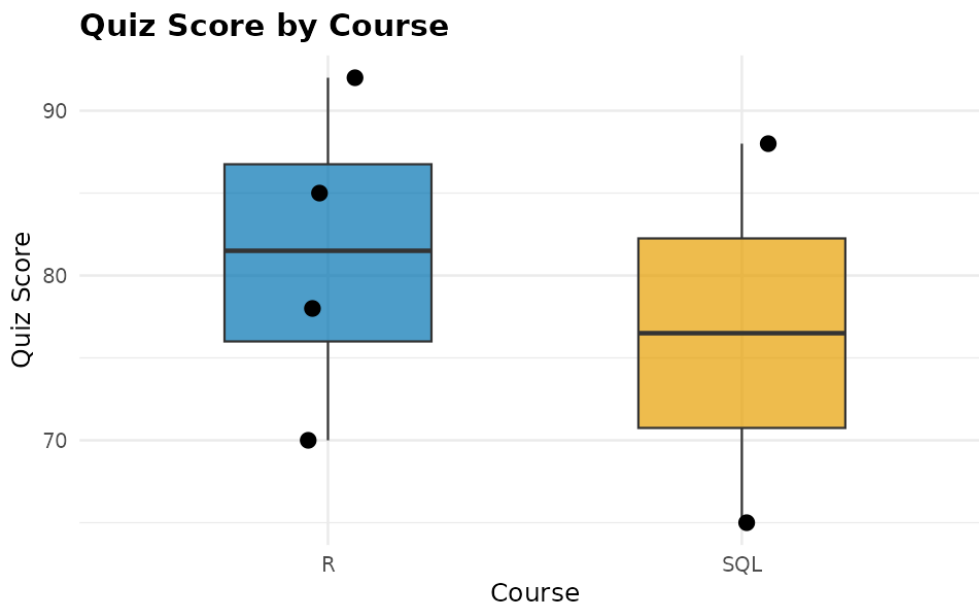


Figure 2: Boxplot of quiz score by course

Inference:

Overall: scores range from 65 to 92, with a mean of 79.67, a median of 81.5 and a standard deviation of 10.6. The histogram has two groups: a lower one (65–70: L03, L05) and a higher one (78–92: the other four students).

By course: R students score higher on average (mean 81.25, median 81.5) than SQL students (mean 76.5). SQL has the wider spread (65 to 88), so its performance is less consistent. L04 (R, 92) is the top performer and L03 (SQL, 65) needs the most support. Only two students take SQL, so the comparison is indicative.

Q2. Scatter Plot – Study_Time vs Quiz_Score with Bubble Size

Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

R Program:

```
# Q2. Scatter plot Study_Time vs Quiz_Score, bubble size = Videos_Watched, transparency
cat("Correlation Study_Time vs Quiz_Score :",
    round(cor(learn$Study_Time, learn$Quiz_Score), 3), "\n")
cat("Correlation Videos_Watched vs Quiz_Score:",
    round(cor(learn$Videos_Watched, learn$Quiz_Score), 3), "\n")
fit <- lm(Quiz_Score ~ Study_Time, data = learn)
cat("Regression: Quiz_Score =", round(coef(fit)[1], 2), "+", round(coef(fit)[2], 2),
    "* Study_Time (R-squared =", round(summary(fit)$r.squared, 3), ")\n")

p_sc <- ggplot(learn, aes(x = Study_Time, y = Quiz_Score)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
    colour = "grey50", linetype = "dashed") +
  geom_point(aes(size = Videos_Watched, colour = Course), alpha = 0.6) +
  geom_text(aes(label = Student_ID), vjust = -1.6, size = 3.3) +
  scale_size(range = c(3, 14), breaks = c(8, 12, 15, 18), name = "Videos Watched") +
  scale_colour_manual(values = c(R = "#0072B2", SQL = "#E69F00")) +
  scale_x_continuous(limits = c(1.5, 5.5)) +
  scale_y_continuous(limits = c(60, 100)) +
  labs(title = "Study Time vs Quiz Score (bubble size = Videos Watched)",
    x = "Study Time (hrs)", y = "Quiz Score", colour = "Course") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
ggsave("s9_q2_scatter.png", p_sc, width = 7.5, height = 4.8, dpi = 150, bg = "white")
```

Console Output:

```
Correlation Study_Time vs Quiz_Score : 0.982
Correlation Videos_Watched vs Quiz_Score: 0.971
Regression: Quiz_Score = 46.75 + 9.32 * Study_Time (R-squared = 0.964 )
```

Output:

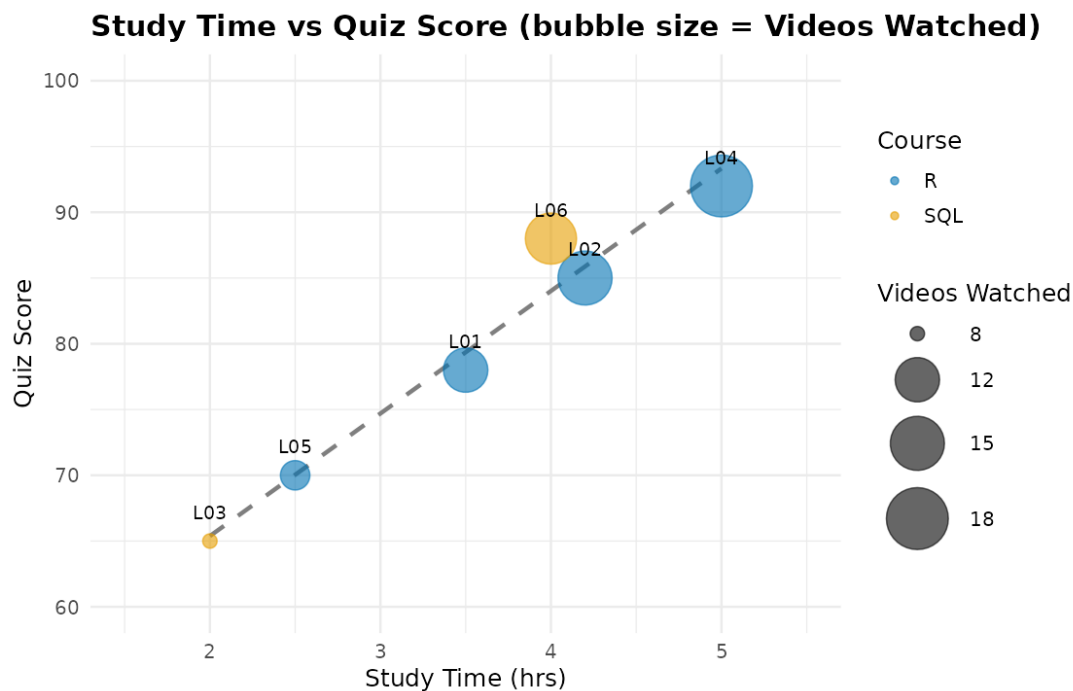


Figure 3: Study time vs quiz score; bubble size = videos watched; alpha = 0.6

Inference:

Learning pattern: there is a very strong positive relationship between study time and quiz score ($r = 0.982$). The regression line shows about **+9.3 marks for every extra hour** of study ($R^2 = 0.964$).

Videos watched follows the same pattern ($r = 0.971$). The largest bubbles (L04 with 18 videos, L02 with 15) sit at the top-right, and the smallest (L03 with 8, L05 with 9) at the bottom-left. Students who engage more, through both study time and videos, score higher. L06 (SQL) scores above the trend line for its study time, which suggests efficient study. Transparency (alpha = 0.6) keeps overlapping bubbles visible.

Q3. Date Conversion, Monthly Average and Moving Average

Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

R Program:

```
# Q3. Convert Login_Date to Date, monthly average quiz score, moving average
learn$Login_Date <- as.Date(learn$Login_Date, format = "%Y-%m-%d")
cat("Class of Login_Date:", class(learn$Login_Date), "\n")
learn$Month <- as.Date(format(learn$Login_Date, "%Y-%m-01"))

monthly <- aggregate(Quiz_Score ~ Month, data = learn, FUN = mean)
names(monthly)[2] <- "Avg_Quiz_Score"
# 2-month trailing moving average
monthly$Moving_Avg <- as.numeric(stats::filter(monthly$Avg_Quiz_Score,
                                                rep(1 / 2, 2), sides = 1))

monthly$Month_Label <- format(monthly$Month, "%b %Y")
print(monthly, row.names = FALSE)

p_line <- ggplot(monthly, aes(x = Month)) +
  geom_line(aes(y = Avg_Quiz_Score, colour = "Monthly average"), linewidth = 1.1) +
  geom_point(aes(y = Avg_Quiz_Score, colour = "Monthly average"), size = 3.5) +
  geom_line(aes(y = Moving_Avg, colour = "2-month moving average"),
            linewidth = 1.1, linetype = "dashed", na.rm = TRUE) +
  geom_point(aes(y = Moving_Avg, colour = "2-month moving average"), size = 3, na.rm = TRUE) +
  geom_text(aes(y = Avg_Quiz_Score, label = Avg_Quiz_Score), vjust = -1.2, size = 3.6) +
  scale_colour_manual(values = c("Monthly average" = "#0072B2",
                                "2-month moving average" = "#D55E00")) +
  scale_x_date(date_labels = "%b %Y", breaks = monthly$Month) +
  scale_y_continuous(limits = c(75, 85)) +
  labs(title = "Average Quiz Score per Month with Moving Average",
       x = "Month", y = "Average Quiz Score", colour = NULL) +
  theme_minimal(base_size = 12) +
```

```
theme(plot.title = element_text(face = "bold"), legend.position = "bottom")
ggsave("s9_q3_monthly.png", p_line, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Class of Login_Date: Date
  Month Avg_Quiz_Score Moving_Avg Month_Label
2025-01-01         81.5         NA      Jan 2025
2025-02-01         78.5        80.00     Feb 2025
2025-03-01         79.0        78.75     Mar 2025
```

Output:

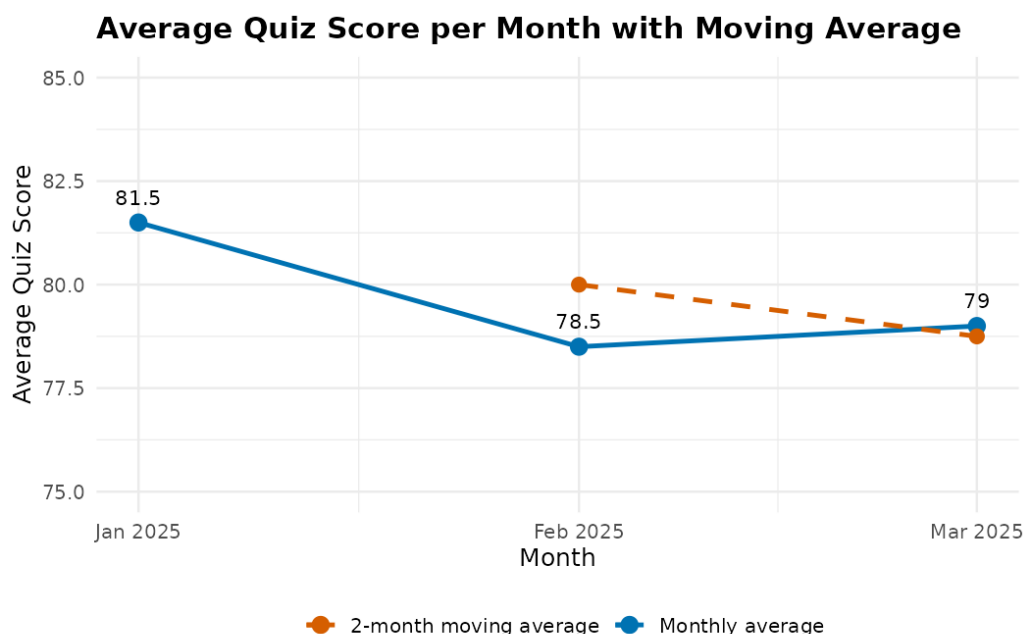


Figure 4: Monthly average quiz score with a 2-month moving average

Inference:

Login_Date was converted from text to the **Date** class with `as.Date()`. The monthly averages are January 81.5, February 78.5 and March 79.0.

Trend: the 2-month moving average falls from 80.0 to 78.75, which shows a **slight downward trend**. Scores dipped in February (pulled down by L03's 65) and recovered only a little in March. With only three months of data, smoothing uses a short window of 2, and the trend should be confirmed with more months.

Q4. Tableau – Bar Chart, Scatter Plot, Filters and Mini Dashboard

Import *Online_Learning_Activity.csv* into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

R Program:

```
# Q4. Dashboard layout preview (Tableau dashboard built from Online_Learning_Activity.csv)
library(patchwork)
course_avg <- aggregate(Quiz_Score ~ Course, data = learn, FUN = mean)
p_bar <- ggplot(course_avg, aes(Course, Quiz_Score, fill = Course)) +
  geom_col(width = 0.5) +
  geom_text(aes(label = round(Quiz_Score, 2)), vjust = -0.5) +
  scale_fill_manual(values = c(R = "#0072B2", SQL = "#E69F00")) +
  scale_y_continuous(limits = c(0, 100)) +
  labs(title = "Average Quiz Score by Course", x = "Course", y = "Avg. Quiz Score") +
  theme_minimal(base_size = 11) + theme(legend.position = "none")
p_sc2 <- ggplot(learn, aes(Study_Time, Quiz_Score, colour = Course, shape = Gender)) +
  geom_point(size = 4) +
  scale_colour_manual(values = c(R = "#0072B2", SQL = "#E69F00")) +
  labs(title = "Study Time vs Quiz Score", x = "Study Time (hrs)", y = "Quiz Score") +
  theme_minimal(base_size = 11)
filters <- grid::textGrob(
  "FILTERS\n\nGender\n[x] Female\n[x] Male\n\nCourse\n[x] R\n[x] SQL",
```

```
x = 0.1, hjust = 0, gp = grid::gpar(fontsize = 11, fontfamily = "mono"))
dash <- (p_cbar | p_sc2 | wrap_elements(full = filters)) +
  plot_layout(widths = c(1, 1.3, 0.45)) +
  plot_annotation(title = "Online Learning Activity - Mini Interactive Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 15)))
ggsave("s9_dashboard.png", dash, width = 12, height = 5, dpi = 130, bg = "white")
cat("Average quiz score by course:\n"); print(course_avg, row.names = FALSE)
```

Console Output:

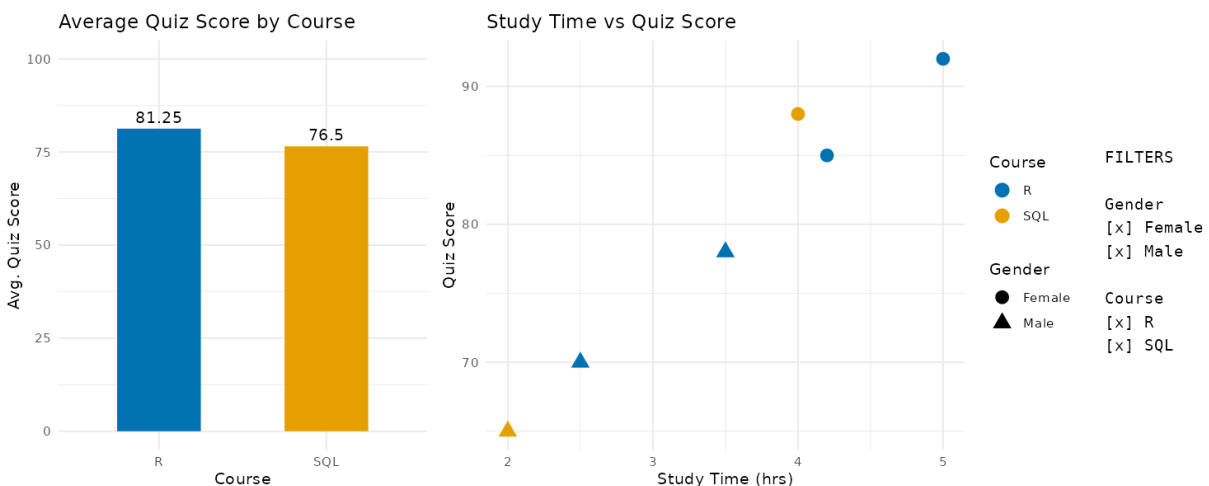
```
Average quiz score by course:
Course Quiz_Score
R      81.25
SQL     76.50
```

Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Online_Learning_Activity.csv**. In the Data Source tab, check that Login_Date is typed as **Date** and Student_ID is a Dimension.
- 2 **Sheet 1 – "Avg Quiz by Course"**: drag Course to Columns and Quiz_Score to Rows. Change the aggregation to **Average** (right-click → Measure → Average). Drag Course to Color and turn on mark labels. Expected values: R = 81.25, SQL = 76.5.
- 3 **Sheet 2 – "Study Time vs Quiz Score"**: drag Study_Time to Columns and Quiz_Score to Rows. Drag **Student_ID to Detail** to show one point per student (or uncheck Analysis → Aggregate Measures). Drag Course to Color, Gender to Shape, and Videos_Watched to Size. Optionally add **Analytics** → **Trend Line**.
- 4 **Filters**: drag Gender and Course to the **Filters** shelf → right-click → **Show Filter**. Then right-click each filter → **Apply to Worksheets** → **All Using This Data Source** so both charts respond together.
- 5 **Dashboard** → **New Dashboard**: place "Avg Quiz by Course" on the left and the scatter plot on the right. Put the Gender and Course filter cards in a vertical container on the right, and add a title.
- 6 **Interactivity**: on the bar chart, select **Use as Filter**. Clicking the R or SQL bar then filters the scatter plot. Save as **.twbx** or publish to Tableau Public.

Dashboard Layout:

Online Learning Activity - Mini Interactive Dashboard



Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

The dashboard lets users select a course or gender and immediately see how average scores and the study-time relationship change. For example, filtering to Female shows the three highest scorers (85, 92, 88).

Result

The online learning dataset was imported into R and analysed with a histogram, boxplot, transparent bubble scatter plot and a moving-average trend line. A filtered mini dashboard was designed in Tableau. Quiz performance is strongly driven by study time and videos watched ($r \approx 0.98$), R students outperform SQL students on average, and the monthly trend is slightly downward.